

Hamidreza Akef

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Research Interests

Molecular programming, chemical reaction networks, DNA computing and nanotechnology, unconventional computation, and noncoding RNA networks.

Education

The University of Texas at Austin

Aug 2023–Present

Ph.D. in Electrical and Computer Engineering

- Advisor: David Soloveichik
- Research area: molecular programming and unconventional computation

The University of Texas at Austin

Aug 2023–Dec 2025

M.S. in Electrical and Computer Engineering

University of Tehran

Sep 2018–Jul 2022

B.S. in Mechanical Engineering

- GPA: 3.96/4.00; ranked 2nd among 129 students

Publications

Amplification at Equilibrium: Structural and Thermodynamic Limitations, and Implementation

H. Akef, C. Y. Sung, A. Vanguri, and D. Soloveichik.

Accepted at the 32nd International Conference on DNA Computing and Molecular Programming (DNA32), 2026.
[arXiv](#) 

Computing and Bounding Equilibrium Concentrations in Athermic Chemical Systems

H. Akef, M. Hhan, and D. Soloveichik.

In *Proceedings of the 31st International Conference on DNA Computing and Molecular Programming (DNA31)*, LIPIcs, 2025. [DOI](#) 

Honors and Awards

NSF Student Travel Grant, DNA32 Conference 2026

DNA31 Excellent Student Paper Award 2025

University of Texas at Austin Graduate School Fellowship 2023–2024

Ranked 2nd among 129 B.S. Mechanical Engineering students, University of Tehran 2018–2022

2nd place, 26th National Mechanics Olympiad of Iran 2021

Ranked 149th of 55,000 in Iran's National University Entrance Examination 2018

Presentations

From Hairballs to Algorithms: Understanding ncRNA Networks through DNA Nanotechnology Design Principles

H. Akef and D. Soloveichik.

Regulatory and Non-Coding RNAs, Cold Spring Harbor Laboratory, Cold Spring Harbor, NY, April 2026.

Research Experience

Graduate Research Assistant

Soloveichik Group, The University of Texas at Austin

Austin, TX

Jan 2024–Present

- Develop theoretical and algorithmic foundations for molecular programming, with an emphasis on equilibrium chemical systems and signal amplification.
- Analyze noncoding RNA networks using design principles from DNA nanotechnology.

Undergraduate Research Assistant

NBIC Startups, Cognitive Science Section

Tehran, Iran

Aug 2022–Jan 2023

- Designed and fabricated a microdrive for reliable chronic multi-tetrode recording and optogenetic experiments in freely behaving rodents.
- Contributed to brain–computer interface research in a multidisciplinary team.

Undergraduate Research Assistant

SEECs Lab, University of Tehran

Tehran, Iran

Jan 2022–Jul 2022

- Designed and fabricated a magnet-based tactile sensor for surgical robotics.

Coordinator and Researcher

Jikal Group, University of Tehran

Tehran, Iran

Jun 2020–Jan 2022

- Coordinated the modeling and fabrication of a wearable device for Parkinsonian tremor suppression.
- The project received a financial award at the 10th International Conference on Acoustics and Vibration in Tehran.

Teaching Experience

Teaching Assistant, Dynamic Systems and Controls Laboratory

Walker Department of Mechanical Engineering, UT Austin

Austin, TX

Aug 2023–May 2024

- Instructed students in Python and Arduino and evaluated course assignments.

Undergraduate Teaching Assistant, Dynamics

School of Mechanical Engineering, University of Tehran

Tehran, Iran

Jul 2020–Jul 2022

- Prepared problem solutions, designed course projects, produced MSC Adams tutorials, and graded MATLAB and Adams assignments.

Undergraduate Teaching Assistant, Physics I

School of Engineering Science, University of Tehran

Tehran, Iran

Sep 2019–Jan 2020

- Evaluated quizzes and homework assignments.

Industry Experience

Software Intern

Sina Robotics and Medical Innovators Co. Ltd.

Tehran, Iran

Jun 2021–Sep 2021

- Contributed to a surgical simulator developed in Unity using C#.

Leadership and Professional Membership

President, Persian Student Society, UT Austin

Jun 2025–Oct 2025

Social Media Coordinator, Persian Student Society, UT Austin

Apr 2024–May 2025

Student Member, International Society for Nanoscale Science, Computation and Engineering (ISNSCE)

2024–2026

Technical Skills

Programming: Python, MATLAB, C#

Tools: Arduino, COMSOL, MSC Adams, SolidWorks, EES